

Decision Tree Practical Work - Explanation

Step 1 - Import Libraries

We imported the required libraries for data manipulation (Pandas), building the Decision Tree model, splitting the dataset, and evaluating the model.

Example:

```
import pandas as pd
from sklearn.tree import DecisionTreeClassifier
```

Step 2 - Load the Dataset

The Titanic dataset was loaded into a Pandas DataFrame.

Example:

```
df = pd.read_csv('Titanic-Dataset.csv')
```

Step 3 - Explore the Dataset

We displayed the first rows, column names, dataset information, statistics, and missing values.

```
df.head()
df.info()
df.isnull().sum()
```

Step 4 - Clean the Dataset

We removed the Cabin column, filled missing Age values with the median, and replaced missing Embarked values with the most frequent value.

```
df.drop('Cabin', axis=1, inplace=True)
df['Age']=df['Age'].fillna(df['Age'].median())
```

Step 5 - Remove Unnecessary Columns

PassengerId, Name and Ticket were removed because they do not help the prediction.

```
df.drop(['PassengerId', 'Name', 'Ticket'], axis=1, inplace=True)
```

Step 6 - Encode Categorical Data

Sex and Embarked were converted into numerical values.

```
df['Sex']=df['Sex'].map({'male':0, 'female':1})
```

Step 7 - Define Features and Target

We separated the input features (X) and the target (y).

```
X=df.drop('Survived',axis=1)
y=df['Survived']
```

Step 8 - Split the Dataset

The dataset was divided into 80% training data and 20% testing data.

```
train_test_split(X,y, test_size=0.2, random_state=42)
```

Step 9 – Create and Train the Model

We created the Decision Tree classifier and trained it using the training data.

```
model=DecisionTreeClassifier(random_state=42)  
model.fit(X_train,y_train)
```

Step 10 – Make Predictions

The trained model predicted passenger survival.

```
y_pred=model.predict(X_test)
```

Step 11 – Evaluate the Model

We evaluated the model using Accuracy, Confusion Matrix, and Classification Report.

```
accuracy_score(y_test,y_pred)  
confusion_matrix(y_test,y_pred)
```

Conclusion

This practical work demonstrated the complete Decision Tree classification workflow on the Titanic dataset, from data preparation to model evaluation.